CALIBRATION CERTIFICATE

Certificate Number	R6161-CAL-20260730-122013
Date of Calibration	2026-07-30
Date of Issue	2026-07-30
Type of Calibration	Calibration (As-Found and As-Left)

Calibration Laboratory

Name	Example Metrology Laboratory
Address	Unit 4, Kelvin Way, Example City EX1 2AB
Accreditation	Not accredited -- example document

Customer

Name	Instrument Workshop
Address	12 Bench Road, Example City EX3 4CD

Instrument Under Test

Description	Advantest Voltage Current Source
Model	R6161
Serial Number	3021015
Firmware	A05
Options	-

Reference Standards Used

Description	Model	Serial No.	Cal. Due	Traceability / Cert. No.
Reference multimeter (HP/Keysight 3458A)	3458A	MY45001234	2027-02-11	NMI, cert 3458-11902

Environmental Conditions

Temperature	23.0 C
Relative Humidity	45 %RH
Procedure	Advantest R6161 performance verification. Each range is generated at the points listed and measured with the reference multimeter shown; the deviation is graded against the manufacturer's published total accuracy specification (Operation Manual section 8.1, table 2), 90 days interval. The current ranges were measured directly, through the reference multimeter's own current terminals. The specification assumes more than one hour of warm-up, an ambient of 23 +/-5 degrees C and relative humidity of 70 % or less. Current ranges are assured within an output compliance of +/-10 V.

As-Found Results (before adjustment)

Parameter	Frequency	Nominal Value	As-Used Value	Measured Value	Deviation	Tolerance (+/-)	Uncertainty (+/-)	Result
1V								
1V +F.S	DC	1 V	1 V	1.0000146 V	0.0000146 V	0.000036 V	0.0000087 V	**PASS**
1V +50%	DC	0.5 V	0.5 V	0.5000081 V	0.0000081 V	0.0000235 V	0.0000047 V	**PASS**
1V Zero	DC	0 V	0 V	0.00000160 V	0.00000160 V	0.000011 V	0.00000072 V	**PASS**
1V -50%	DC	-0.5 V	-0.5 V	-0.5000049 V	-0.0000049 V	0.0000235 V	0.0000047 V	**PASS**
1V -F.S	DC	-1 V	-1 V	-1.0000114 V	-0.0000114 V	0.000036 V	0.0000087 V	**PASS**
10V								
10V +F.S	DC	10 V	10 V	10.000146 V	0.000146 V	0.00032 V	0.000084 V	**PASS**
10V +50%	DC	5 V	5 V	5.000081 V	0.000081 V	0.000195 V	0.000044 V	**PASS**
10V Zero	DC	0 V	0 V	0.0000160 V	0.0000160 V	0.00007 V	0.0000049 V	**PASS**
10V -50%	DC	-5 V	-5 V	-5.000049 V	-0.000049 V	0.000195 V	0.000044 V	**PASS**
10V -F.S	DC	-10 V	-10 V	-10.000114 V	-0.000114 V	0.00032 V	0.000084 V	**PASS**
100V								
100V +F.S	DC	100 V	100 V	100.00146 V	0.00146 V	0.0032 V	0.00084 V	**PASS**
100V +50%	DC	50 V	50 V	50.00081 V	0.00081 V	0.00195 V	0.00044 V	**PASS**
100V Zero	DC	0 V	0 V	0.000160 V	0.000160 V	0.0007 V	0.000047 V	**PASS**
100V -50%	DC	-50 V	-50 V	-50.00049 V	-0.00049 V	0.00195 V	0.00044 V	**PASS**
100V -F.S	DC	-100 V	-100 V	-100.00114 V	-0.00114 V	0.0032 V	0.00084 V	**PASS**
1000V								
1000V +F.S	DC	1,000 V	1,000 V	1,000.0146 V	0.0146 V	0.037 V	0.0084 V	**PASS**
1000V +50%	DC	500 V	500 V	500.0081 V	0.0081 V	0.022 V	0.0044 V	**PASS**
1000V Zero	DC	0 V	0 V	0.00160 V	0.00160 V	0.007 V	0.00047 V	**PASS**
1000V -50%	DC	-500 V	-500 V	-500.0049 V	-0.0049 V	0.022 V	0.0044 V	**PASS**
1000V -F.S	DC	-1,000 V	-1,000 V	-1,000.0114 V	-0.0114 V	0.037 V	0.0084 V	**PASS**
10mV DIV								
10mV +F.S	DC	10 mV	10 mV	10.00435 mV	0.00435 mV	0.0029 mV	0.00075 mV	**FAIL**
10mV +50%	DC	5 mV	5 mV	5.00218 mV	0.00218 mV	0.0026 mV	0.00070 mV	**FAIL**
10mV Zero	DC	0 mV	0 mV	0.00002 mV	0.00002 mV	0.0023 mV	0.00065 mV	**PASS**
10mV -50%	DC	-5 mV	-5 mV	-5.00215 mV	-0.00215 mV	0.0026 mV	0.00070 mV	**FAIL**
10mV -F.S	DC	-10 mV	-10 mV	-10.00431 mV	-0.00431 mV	0.0029 mV	0.00075 mV	**FAIL**
100mV DIV								
100mV +F.S	DC	100 mV	100 mV	100.0105 mV	0.0105 mV	0.007 mV	0.0017 mV	**FAIL**
100mV +50%	DC	50 mV	50 mV	50.0053 mV	0.0053 mV	0.00475 mV	0.0012 mV	**FAIL**
100mV Zero	DC	0 mV	0 mV	0.00016 mV	0.00016 mV	0.0025 mV	0.00065 mV	**PASS**
100mV -50%	DC	-50 mV	-50 mV	-50.0050 mV	-0.0050 mV	0.00475 mV	0.0012 mV	**FAIL**
100mV -F.S	DC	-100 mV	-100 mV	-100.0101 mV	-0.0101 mV	0.007 mV	0.0017 mV	**FAIL**
1000mV DIV								
1000mV +F.S	DC	1 V	1 V	1.0000246 V	0.0000246 V	0.000043 V	0.0000085 V	**PASS**
1000mV +50%	DC	0.5 V	0.5 V	0.5000131 V	0.0000131 V	0.0000255 V	0.0000045 V	**PASS**
1000mV Zero	DC	0 V	0 V	0.00000160 V	0.00000160 V	0.000008 V	0.00000061 V	**PASS**
1000mV -50%	DC	-0.5 V	-0.5 V	-0.5000099 V	-0.0000099 V	0.0000255 V	0.0000045 V	**PASS**
1000mV -F.S	DC	-1 V	-1 V	-1.0000214 V	-0.0000214 V	0.000043 V	0.0000085 V	**PASS**
1mA								
1mA +F.S	DC	1 mA	1 mA	1.000015 mA	0.000015 mA	0.000069 mA	0.000027 mA	**PASS**
1mA +50%	DC	0.5 mA	0.5 mA	0.500008 mA	0.000008 mA	0.000039 mA	0.000017 mA	**PASS**
1mA Zero	DC	0 mA	0 mA	0.0000016 mA	0.0000016 mA	0.000009 mA	0.0000070 mA	**PASS**
1mA -50%	DC	-0.5 mA	-0.5 mA	-0.500005 mA	-0.000005 mA	0.000039 mA	0.000017 mA	**PASS**
1mA -F.S	DC	-1 mA	-1 mA	-1.000011 mA	-0.000011 mA	0.000069 mA	0.000027 mA	**PASS**

Parameter	Frequency	Nominal Value	As-Used Value	Measured Value	Deviation	Tolerance (+/-)	Uncertainty (+/-)	Result
10mA								
10mA +F.S	DC	10 mA	10 mA	10.00015 mA	0.00015 mA	0.00054 mA	0.00026 mA	**PASS**
10mA +50%	DC	5 mA	5 mA	5.00008 mA	0.00008 mA	0.000315 mA	0.00016 mA	**PASS**
10mA Zero	DC	0 mA	0 mA	0.000016 mA	0.000016 mA	0.00009 mA	0.000061 mA	**PASS**
10mA -50%	DC	-5 mA	-5 mA	-5.00005 mA	-0.00005 mA	0.000315 mA	0.00016 mA	**PASS**
10mA -F.S	DC	-10 mA	-10 mA	-10.00011 mA	-0.00011 mA	0.00054 mA	0.00026 mA	**PASS**
100mA								
100mA +F.S	DC	100 mA	100 mA	100.0015 mA	0.0015 mA	0.0054 mA	0.0041 mA	**FAIL**
100mA +50%	DC	50 mA	50 mA	50.0008 mA	0.0008 mA	0.00315 mA	0.0024 mA	**FAIL**
100mA Zero	DC	0 mA	0 mA	0.00016 mA	0.00016 mA	0.0009 mA	0.00060 mA	**PASS**
100mA -50%	DC	-50 mA	-50 mA	-50.0005 mA	-0.0005 mA	0.00315 mA	0.0024 mA	**PASS**
100mA -F.S	DC	-100 mA	-100 mA	-100.0011 mA	-0.0011 mA	0.0054 mA	0.0041 mA	**PASS**

As-Left Results (after adjustment)

Parameter	Frequency	Nominal Value	As-Used Value	Measured Value	Deviation	Tolerance (+/-)	Uncertainty (+/-)	Result
1V								
1V +F.S	DC	1 V	1 V	1.0000038 V	0.0000038 V	0.000036 V	0.0000087 V	**PASS**
1V +50%	DC	0.5 V	0.5 V	0.5000020 V	0.0000020 V	0.0000235 V	0.0000047 V	**PASS**
1V Zero	DC	0 V	0 V	0.00000030 V	0.00000030 V	0.000011 V	0.00000072 V	**PASS**
1V -50%	DC	-0.5 V	-0.5 V	-0.5000015 V	-0.0000015 V	0.0000235 V	0.0000047 V	**PASS**
1V -F.S	DC	-1 V	-1 V	-1.0000032 V	-0.0000032 V	0.000036 V	0.0000087 V	**PASS**
10V								
10V +F.S	DC	10 V	10 V	10.000038 V	0.000038 V	0.00032 V	0.000084 V	**PASS**
10V +50%	DC	5 V	5 V	5.000021 V	0.000021 V	0.000195 V	0.000044 V	**PASS**
10V Zero	DC	0 V	0 V	0.0000030 V	0.0000030 V	0.00007 V	0.0000049 V	**PASS**
10V -50%	DC	-5 V	-5 V	-5.000014 V	-0.000014 V	0.000195 V	0.000044 V	**PASS**
10V -F.S	DC	-10 V	-10 V	-10.000032 V	-0.000032 V	0.00032 V	0.000084 V	**PASS**
100V								
100V +F.S	DC	100 V	100 V	100.00038 V	0.00038 V	0.0032 V	0.00084 V	**PASS**
100V +50%	DC	50 V	50 V	50.00021 V	0.00021 V	0.00195 V	0.00044 V	**PASS**
100V Zero	DC	0 V	0 V	0.000030 V	0.000030 V	0.0007 V	0.000047 V	**PASS**
100V -50%	DC	-50 V	-50 V	-50.00014 V	-0.00014 V	0.00195 V	0.00044 V	**PASS**
100V -F.S	DC	-100 V	-100 V	-100.00032 V	-0.00032 V	0.0032 V	0.00084 V	**PASS**
1000V								
1000V +F.S	DC	1,000 V	1,000 V	1,000.0038 V	0.0038 V	0.037 V	0.0084 V	**PASS**
1000V +50%	DC	500 V	500 V	500.0020 V	0.0020 V	0.022 V	0.0044 V	**PASS**
1000V Zero	DC	0 V	0 V	0.00030 V	0.00030 V	0.007 V	0.00047 V	**PASS**
1000V -50%	DC	-500 V	-500 V	-500.0015 V	-0.0015 V	0.022 V	0.0044 V	**PASS**
1000V -F.S	DC	-1,000 V	-1,000 V	-1,000.0032 V	-0.0032 V	0.037 V	0.0084 V	**PASS**
10mV DIV								
10mV +F.S	DC	10 mV	10 mV	10.00004 mV	0.00004 mV	0.0029 mV	0.00075 mV	**PASS**
10mV +50%	DC	5 mV	5 mV	5.00002 mV	0.00002 mV	0.0026 mV	0.00070 mV	**PASS**
10mV Zero	DC	0 mV	0 mV	0.00000 mV	0.00000 mV	0.0023 mV	0.00065 mV	**PASS**
10mV -50%	DC	-5 mV	-5 mV	-5.00001 mV	-0.00001 mV	0.0026 mV	0.00070 mV	**PASS**
10mV -F.S	DC	-10 mV	-10 mV	-10.00003 mV	-0.00003 mV	0.0029 mV	0.00075 mV	**PASS**
100mV DIV								
100mV +F.S	DC	100 mV	100 mV	100.0004 mV	0.0004 mV	0.007 mV	0.0017 mV	**PASS**
100mV +50%	DC	50 mV	50 mV	50.0002 mV	0.0002 mV	0.00475 mV	0.0012 mV	**PASS**
100mV Zero	DC	0 mV	0 mV	0.00003 mV	0.00003 mV	0.0025 mV	0.00065 mV	**PASS**
100mV -50%	DC	-50 mV	-50 mV	-50.0001 mV	-0.0001 mV	0.00475 mV	0.0012 mV	**PASS**
100mV -F.S	DC	-100 mV	-100 mV	-100.0003 mV	-0.0003 mV	0.007 mV	0.0017 mV	**PASS**
1000mV DIV								
1000mV +F.S	DC	1 V	1 V	1.0000038 V	0.0000038 V	0.000043 V	0.0000085 V	**PASS**
1000mV +50%	DC	0.5 V	0.5 V	0.5000020 V	0.0000020 V	0.0000255 V	0.0000045 V	**PASS**
1000mV Zero	DC	0 V	0 V	0.00000030 V	0.00000030 V	0.000008 V	0.00000061 V	**PASS**
1000mV -50%	DC	-0.5 V	-0.5 V	-0.5000015 V	-0.0000015 V	0.0000255 V	0.0000045 V	**PASS**
1000mV -F.S	DC	-1 V	-1 V	-1.0000032 V	-0.0000032 V	0.000043 V	0.0000085 V	**PASS**
1mA								
1mA +F.S	DC	1 mA	1 mA	1.000004 mA	0.000004 mA	0.000069 mA	0.000027 mA	**PASS**
1mA +50%	DC	0.5 mA	0.5 mA	0.500002 mA	0.000002 mA	0.000039 mA	0.000017 mA	**PASS**
1mA Zero	DC	0 mA	0 mA	0.0000003 mA	0.0000003 mA	0.000009 mA	0.0000070 mA	**PASS**
1mA -50%	DC	-0.5 mA	-0.5 mA	-0.500001 mA	-0.000001 mA	0.000039 mA	0.000017 mA	**PASS**
1mA -F.S	DC	-1 mA	-1 mA	-1.000003 mA	-0.000003 mA	0.000069 mA	0.000027 mA	**PASS**

Parameter	Frequency	Nominal Value	As-Used Value	Measured Value	Deviation	Tolerance (+/-)	Uncertainty (+/-)	Result
10mA								
10mA +F.S	DC	10 mA	10 mA	10.00004 mA	0.00004 mA	0.00054 mA	0.00026 mA	**PASS**
10mA +50%	DC	5 mA	5 mA	5.00002 mA	0.00002 mA	0.000315 mA	0.00016 mA	**PASS**
10mA Zero	DC	0 mA	0 mA	0.000003 mA	0.000003 mA	0.00009 mA	0.000061 mA	**PASS**
10mA -50%	DC	-5 mA	-5 mA	-5.00001 mA	-0.00001 mA	0.000315 mA	0.00016 mA	**PASS**
10mA -F.S	DC	-10 mA	-10 mA	-10.00003 mA	-0.00003 mA	0.00054 mA	0.00026 mA	**PASS**
100mA								
100mA +F.S	DC	100 mA	100 mA	100.0004 mA	0.0004 mA	0.0054 mA	0.0041 mA	**PASS**
100mA +50%	DC	50 mA	50 mA	50.0002 mA	0.0002 mA	0.00315 mA	0.0024 mA	**PASS**
100mA Zero	DC	0 mA	0 mA	0.00003 mA	0.00003 mA	0.0009 mA	0.00060 mA	**PASS**
100mA -50%	DC	-50 mA	-50 mA	-50.0001 mA	-0.0001 mA	0.00315 mA	0.0024 mA	**PASS**
100mA -F.S	DC	-100 mA	-100 mA	-100.0003 mA	-0.0003 mA	0.0054 mA	0.0041 mA	**PASS**

Overall Result: PASS

Pass/Fail is assessed against the manufacturer's published total accuracy specification for the Advantest R6161 (Operation Manual section 8.1, table 2), 90 days interval: +/-(% of setting + range error). Total accuracy includes the traceability of the external standards, calibration error, stability, temperature coefficient, long term stability, linearity and noise ripple (DC to 1 Hz); line and load regulation are excluded.

Statement of Measurement Uncertainty

The reported expanded uncertainty of measurement is stated as the standard uncertainty of the measurement multiplied by a coverage factor k = 2, providing a level of confidence of approximately 95 %. It is evaluated from the laboratory's own budget for the reference standards used, applied at each point's reading and range. The contributions to each budget are summed arithmetically, as the terms of a specification limit. The evaluation includes, per point, the observed spread between repeated readings, combined in quadrature. Conformity is assessed against the manufacturer's published accuracy specification reduced by the reported expanded uncertainty, so that a result too close to the limit to resolve is reported as a failure (guarded acceptance).

Statement of Traceability

The measurements reported in this certificate are traceable to national measurement standards through the reference standards listed above.

Calibrated by: A. Technician	Approved by: Q. Manager
Signature: _____	Signature: _____
Date: _____	Date: _____

This certificate relates only to the item(s) calibrated and shall not be reproduced other than in full without the written approval of the issuing laboratory. This calibration was performed using reverse-engineered, non-factory software and is provided for engineering use.